Advanced Data Science Roadmap

Step-by-Step Plan to Become a Professional Data Scientist

Phase 1: Review & Strengthen the Basics (2-3 weeks)

Key Topics:

- Python (data structures, OOP, file I/O)
- NumPy, Pandas, Matplotlib, Seaborn
- SQL (joins, group by, subqueries)
- Scikit-learn: basic models

Practice:

- HackerRank (SQL + Python)
- LeetCode (Easy)
- Kaggle Titanic project

Phase 2: Intermediate ML & Projects (3-4 weeks)

Topics:

- Feature engineering, model evaluation, cross-validation
- Hyperparameter tuning: GridSearchCV, RandomSearch

Projects:

- House price prediction
- Customer churn detection
- Sentiment analysis

Resources:

- "Hands-On ML with Scikit-Learn, Keras & TF"
- Kaggle's Intermediate ML course

Phase 3: Deep Dive into Deep Learning (4-6 weeks)

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Topics:

- Neural networks, CNNs, RNNs, LSTMs
- TensorFlow or PyTorch

Projects:

- MNIST digit recognition
- Text sentiment classifier
- Image classifier

Resources:

- DeepLearning.AI TensorFlow Specialization
- PyTorch tutorials

Phase 4: Special Topics & Tools (4 weeks)

Topics:

- Time Series Forecasting, NLP
- Streamlit, Flask, Docker, Spark, Cloud (AWS/GCP)

Resources:

- Coursera NLP Specialization
- Kaggle Time Series Course

Phase 5: Career Prep & Real-World Exposure (4+ weeks)

Tasks:

- GitHub portfolio with 3-5 projects
- Technical blogging (Medium/Hashnode)
- Kaggle competitions, apply for internships

Learn:

- Behavioral & technical interview prep

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- SQL + ML case studies

Weekly Schedule Example

Mon-Wed: Learn new topics (videos, articles)

Thu-Fri: Practice (coding, notebooks)

Sat: Project work

Sun: Revise, GitHub/blog updates